BLACK HOLE PHYSICS

A loud quasi-periodic oscillation after a star is disrupted by a massive black hole

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The tidal forces close to massive black holes can rip apart stars that come too close to them. As the resulting stellar debris spirals toward the black hole, the debris heats up and emits x-rays. We report observations of a stable 131-second x-ray quasi-periodic oscillation from the tidal disruption event ASASSN-14li. Assuming the black hole mass indicated by host galaxy scaling relations, these observations imply that the periodicity originates from close to the event horizon and that the black hole is rapidly spinning. Our findings demonstrate that tidal disruption events can generate quasi-periodic oscillations that encode information about the physical properties of their black holes.

Almost all massive galaxies are thought to harbor a massive black hole (MBH) [with a mass of $\gtrsim 10^4$ solar masses ($M_\odot$)] at their centers (*1*), yet most of the MBHs are inactive and do not produce any observable electromagnetic radiation. However, once every ~10^4 to 10^5 years, a star is predicted to pass near enough to the black hole (BH) to be disrupted by the BH's gravitational forces (*2*–*4*). Such episodes, known as tidal disruption events (TDEs) (*5*), trigger accretion of the debris onto quiescent BHs and provide a brief period of activity. This creates an opportunity to measure the two properties that characterize BHs: mass and spin. Empirical scaling laws can be used to infer BH masses, for example, by using host galaxy properties (*6*), but the spins of MBHs have been difficult to constrain. This is because the effects of spin predicted by Einstein's general theory of relativity are negligible except in the immediate vicinity of BHs, typically within a few gravitational radii (*7*). One gravitational radius is $R_g = GM/c^2$, where G, M, and c are the gravitational constant, the BH mass, and the speed of light, respectively. Measuring BH spins requires observations of radiation from the innermost regions of the accretion flow, where gravity is strong. Theoretical models of TDEs predict that shortly after the disruption, a fraction of the stellar debris settles into a hot inner disk with peak thermal emission in the soft–x-ray or extreme–ultraviolet (UV) range (*8*). Identification of such disk-dominated or x-ray–bright TDEs could be used to determine MBH spins.

The transient event ASASSN-14li (right ascension, 12 hours 48 min 15.23 s; declination, 17°46′ 26.22″; J2000.0 equinox) was detected by the All-Sky Automated Survey for Supernovae (ASASSN) on 22 November 2014 (*3*). It exhibited most of the properties of previously known TDEs: a spatial position consistent with the host galaxy's center [within 160 pc (*3*)]; a luminosity declining in time with a power law index of $^5/_3$ (*9*), as expected for a TDE (*10*); and a blue optical spectrum with broad hydrogen and helium emission lines and a constant optical color, unlike that of an ordinary supernova (*3*). ASASSN-14li also produced x-rays (*9*) and a radio synchrotron flare (*11*, *12*).

The masses of central MBHs are known to correlate with the properties of their host galaxies (*6*, *13*). The velocity dispersion of stars in the inner bulges of galaxies (σ_{vel}) is correlated with the BH mass (M), and this correlation is commonly referred to as the M-σ_{vel} relation (*6*, *13*). The total stellar mass in the bulge and the optical luminosity of the host galaxy are also known to correlate with the BH mass (*13*). These empirical relations indicate that the BH in ASASSN-14li has a mass in the range of $10^{5.8}$ to $10^{7.1}$ $M_\odot$ (*3*, *12*, *14*). This range is consistent with the BH mass derived independently from physical modeling of ASASSN-14li's multiwavelength light curves (*9*). The observed x-ray energy spectrum is blackbody-like (thermal) (*9*, *15*, *16*), with peak 0.3- to 1.0-keV luminosity of a few times 10^{43} erg/s (Fig. 1). The inferred size of the thermal x-ray–emitting region (~10^{12} cm) is only a few gravitational radii (*9*) and remains roughly constant with time (*9*, *15*). This suggests that x-rays from ASASSN-14li originate from an inner accretion flow close to the BH.

In stellar-mass BHs, a sudden onset of accretion often excites quasi-periodic oscillations (QPOs) in the x-ray flux (*17*). In instances where the x-ray emission is dominated by the accretion disk, observed QPO frequencies have been used to measure the BH spins (*18*, *19*). We searched for a stable QPO in the soft–x-ray–band (0.3- to 1.0-keV) observations of ASASSN-14li by combining publicly available data from the X-ray Multi-Mirror Mission (XMM-Newton) and Chandra X-ray Observatory space telescopes. We extracted the average power density spectrum (PDS) from data taken at six epochs during the 450 days after ASASSN-14li's discovery (Fig. 1). The combined x-ray PDS shows a feature at 7.65 ± 0.4 mHz (131 s; coherence, Q = centroid frequency/QPO's width = 16 ± 6), as shown in Fig. 2. The highest bin in the QPO is statistically significant at the 4.8σ level (where σ represents statistical significance expressed in multiples of standard deviation) for a search at all frequencies (trials) below 0.5 Hz (Fig. 2A) under the white-noise hypothesis (noise variations are independent of the time scale). Although the data are consistent with white noise, by assuming the most extreme red noise allowed by the data (i.e., that noise scales inversely with frequency) we derive a conservative lower limit on the statistical significance (false alarm probability) of the highest QPO bin to be 3.9σ (or 10^{-4}) (*20*).

The QPO is independently detected in the XMM-Newton and Chandra datasets with significance of ≈4σ and ≳2.6σ, respectively, for a search including all frequencies (trials) below 0.5 Hz (*20*) (fig. S9). We estimated the QPO's fractional root-mean-squared (rms) amplitude during the last XMM-Newton epoch to be 4 ± 1% (Fig. 1) (*20*). Because the source was bright and the instrument readout was not fast enough in the first four XMM-Newton observations, the data were piled up (*20*). Thus, similar measurements could not be made for epochs X1 to X4. The Chandra observation was made roughly 420 days after the discovery, by which time ASASSN-14li's flux had declined by ≈10, reducing the pileup (*20*). The QPO's fractional rms amplitude in Chandra data was 59 ± 11% (Fig. 3) (*20*). This suggests that between epochs X5 and C1, separated by ~50 days, the fractional rms amplitude of the QPO increased by at least an order of magnitude. After establishing the QPO at 7.65 mHz, we also constructed an average x-ray (0.3- to 1.0-keV) PDS from observations taken by the Neil Gehrels Swift Observatory. The strongest feature in the average Swift PDS is at 7.0 ± 0.5 mHz, consistent with the QPO detected in the XMM-Newton and Chandra datasets (Fig. 2B).

Plotting the Chandra data in imaging mode shows only a single x-ray point source spatially coincident with the galaxy LEDA 043234 (fig. S5). This demonstrates that the QPO does not originate from a nearby contaminating source. The QPO is detected by three different x-ray detectors, establishing that it is not an instrumental artifact but is associated with ASASSN-14li. Movie S1 shows that the QPO signal improves gradually as more power spectra are averaged, implying that the QPO does not originate from a single epoch observation but is present throughout

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at least the first 450 days of the event. The average Swift PDS from data acquired over 500 days, the Chandra PDS from roughly day 420, and the average XMM-Newton PDS all show QPOs at a consistent frequency throughout the first 450 days of the outburst. This implies that the QPO is stable for 3×10^5 cycles (≈450 days/131 s). Whereas the stability and coherence of the QPO are similar to those of the QPOs of stellar-mass BHs in a disk-dominated state, the modulation amplitude of >50% (Fig. 3) is higher [e.g., (*21*)].

An alternative scenario in which the oscillation might be a neutron star pulsation is unlikely for multiple reasons: the large x-ray, optical-UV, and radio photospheric sizes (*3*, *15*, *16*, *22*); the high bolometric luminosity (*15*, *16*); and the very soft x-ray spectrum (*9*, *15*). In general, the multiwavelength properties of ASASSN-14li are similar to those of many previously known TDEs and unlike those of any known neutron star outburst (see supplementary text).

Assuming ASASSN-14li's BH mass range implied from standard host galaxy scaling relations, we compared the 7.65-mHz QPO frequency with the five possible frequencies of motion of a test particle orbiting a spinning BH (*7*, *19*). The five frequencies are determined by the BH's mass and spin and the radial distance of the emitting region (supplementary text). In disk-dominated stellar-mass BHs, the inner edges of the accretion disks extend to a constant radius, for a wide range in accretion rates [e.g., (*23*)]. The natural inner radius predicted by general relativity is the innermost stable circular orbit (ISCO), which depends on BH spin. Because ASASSN-14li appears to be disk dominated, we started our frequency comparison by using the ISCO for the radial distance (Fig. 4). Even at this closest possible location, the only possible solutions are those with a rapidly spinning BH. A lower limit on the BH's dimensionless spin parameter ($a^* = Jc/GM^2$, where J is BH's angular momentum) can be calculated from the BH spin-versus-mass contours shown in Fig. 4. This corresponds to the intersection of the BH mass lower limit and the fastest frequency, which at any given radius is the Keplerian frequency. This implies that ASASSN-14li's spin parameter is greater than 0.7 (Fig. 4). Placing the test particle at any larger radius would only shift this limit to higher spin values. At any given radius, as the other four frequencies (Fig. 4 and supplementary text) are below the Keplerian value, associating the QPO with them would again shift the spin limit to higher values.

If we ignore frequencies higher than the azimuthal (Keplerian) frequency (but see below), then we can interpret Fig. 4 as showing a lower limit on the spin [e.g., (*24*)] of the MBH that caused the TDE. Alternatively, we can interpret the figure as showing an upper limit of $2 \times 10^6 M_\odot$ on the BH mass, for a maximum astrophysically plausible spin of $a^* = 0.998$ (*25*). The maximal spin comes from the conjecture that naked singularities (such as BHs with $a^* > 1$) are not allowed to exist in nature (*26*)

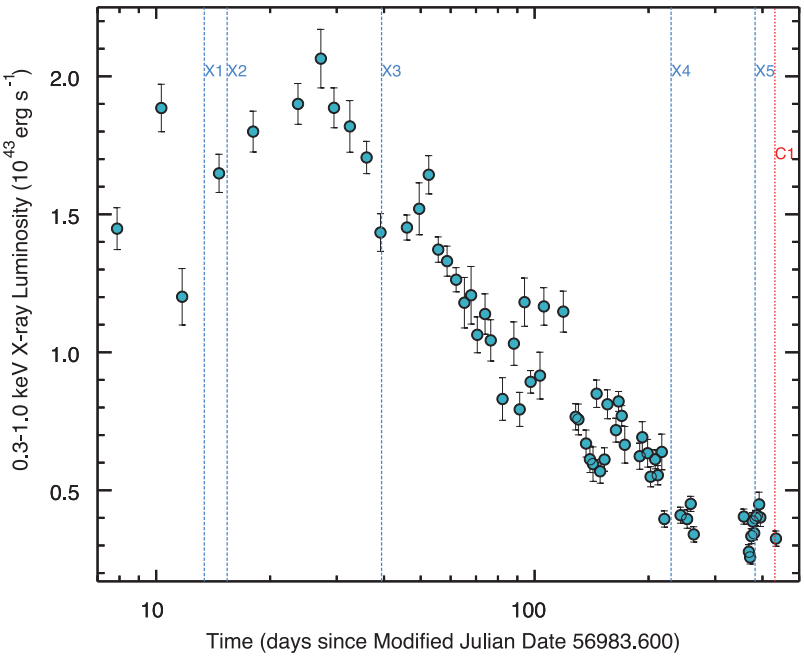


Fig. 1. ASASSN-14li's long-term x-ray light curve. The data were taken with Swift [and corrected for pileup (*20*)]. The dashed vertical lines represent the five epochs of XMM-Newton observations (blue, labeled X1 to X5) and one epoch of Chandra observation (red, labeled C1).

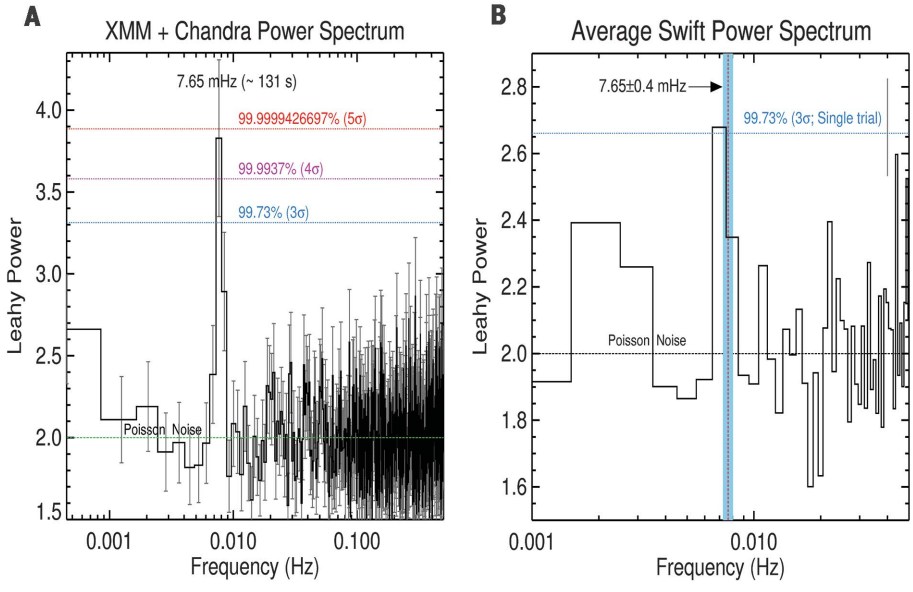


Fig. 2. X-ray power spectra for ASASSN-14li, showing a QPO at 7.65 mHz. (**A**) The average x-ray PDS from eight continuous 10,000-s light curves taken with XMM-Newton and Chandra. The frequency resolution is 0.8 mHz. The strongest feature in the power spectrum lies at a frequency of 7.65 ± 0.4 mHz (≈131 s). The horizontal blue, magenta, and red lines represent the 3, 4, and 5σ white-noise statistical thresholds. The data surrounding the QPO feature are consistent with white noise (*20*), but we also estimated the QPO significance under red noise, finding that its highest bin is significant at at least the 3.9σ level (*20*). Uncertainties of ±1σ are shown with gray error bars. Figure S9 shows the XMM-Newton and Chandra data separately. (**B**) Average Swift PDS from 85 continuous 1000-s light curves with a frequency resolution of 1 mHz. The blue horizontal line shows the 3σ threshold for a single trial search at 7.65 mHz. The highest peak in the power spectrum is at 7.0 ± 0.5 mHz, consistent with the XMM-Newton and Chandra power spectra (fig. S9).

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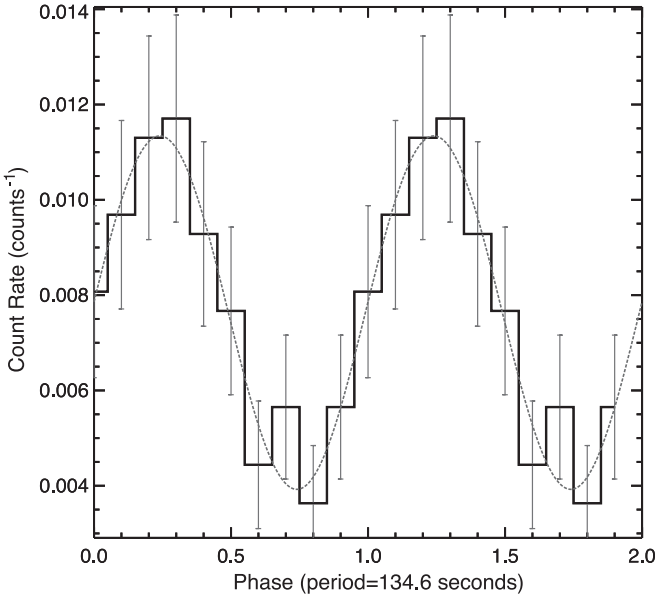


Fig. 3. ASASSN-14li's folded x-ray light curve from Chandra data. The fold period during epoch C1 was estimated by oversampling the light curve (*20*) to be 134.6 ± 0.1 s (or 7.43 ± 0.006 mHz). The best-fitting sinusoidal curve (dashed line) implies a fractional amplitude of 35 ± 8%, consistent (within the 90% confidence limits) with the estimate from the PDS (fig. S9). The zero phase is arbitrary, and two cycles are shown for clarity. Uncertainties of ±1σ are shown as error bars. Figures S10 and S11 show the folded XMM-Newton light curves and the evolution of the QPO's rms amplitude, respectively.

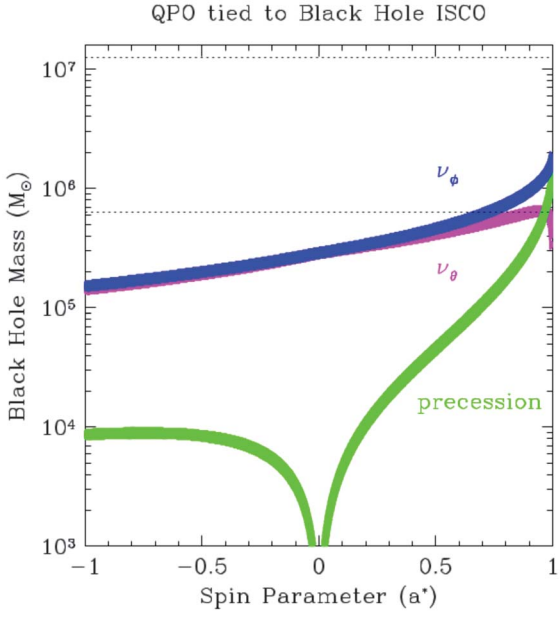


Fig. 4. BH dimensionless-spin-parameter–versus–mass contours. Spin-versus-mass contours determined by assuming that the 7.65-mHz QPO is associated with any of three particle frequencies—Keplerian frequency (ν_ϕ) (blue), vertical epicyclic frequency (ν_θ) (magenta), and Lense-Thirring precession ($\nu_\phi - \nu_\theta$) (green)—at the ISCO, where the radial epicyclic frequency (ν_r) is zero and the periastron precession frequency ($\nu_\phi - \nu_r$) is thus equal to the Keplerian frequency (*20*). The widths of these contours reflect the QPO's width of 0.7 mHz (upper limit). The dashed lines show ASASSN-14li's BH mass range ($10^{5.8}$ to $10^{7.1}\,M_\odot$) estimated from its host galaxy scaling relations. Within this mass range, the only formal solutions are the ones that require the BH spin parameter to be greater than 0.7.

and the reality that countertorques from radiation absorbed into the BH limit the growth of a^* to 0.998 (*25*).

It is possible that ASASSN-14li's host galaxy and the disrupting BH do not obey the empirical scaling laws (*27*) and that instead the BH mass is below a value of a few times $10^5 M_\odot$. If so, then the BH could have a moderate spin, but this would imply that the BH is an intermediate-mass BH, representing a class of objects whose existence has been controversial [e.g., see (*28*–*30*)].

The QPO has a higher dimensionless frequency than those measured from stellar-mass BHs (*17*): QPO frequency/$(c^3/GM) > 0.024$, where we have used the lower limit of the estimated BH mass range (*14*). In stellar-mass BHs, the dimensionless QPO frequencies are $\gtrsim$0.01 (*17*). This implies that the radiating material producing the QPO is located close to the BH's event horizon and rules out alternative models for x-ray radiation that require an emitting region far away from the BH. The physical mechanism that produced the QPO remains unclear (supplementary text).

The QPO in ASASSN-14li has further differences from those arising from stellar-mass BHs. The high-frequency QPOs (with frequencies of a few hundred hertz) of accreting stellar-mass BHs are seen only in hard x-rays (>2 keV) (*17*) and not in disk-dominated states (*21*), whereas ASASSN-14li's energy spectrum is very soft (*9*). The rapid rise in the QPO's rms amplitude is also uncharacteristic of stellar-mass BHs. ASASSN-14li's QPO may represent a different disk oscillation mode from other systems, and thus it may not be valid to compare it directly with known QPOs of stellar-mass BHs.

A quasi-periodicity (at ≈200 s) was previously reported from the TDE SwiftJ164449.3+573451 (SwJ1644+57) (*31*). However, SwJ1644+57 is an atypical TDE in which the entire electromagnetic radiation was dominated by a jet directly pointing along our line of sight [e.g., (*32*)]. Radio follow-up indicates that only a small fraction of thermal TDEs launch collimated jets (*33*), and only a small fraction of such jets would align with our line of sight. SwJ1644+57's periodicity had an amplitude ~$^1/_{15}$ that of ASASSN-14li's and was present only for a short duration of at most a few weeks after its discovery.

High-frequency x-ray QPOs originate from the strong gravity regime in the immediate vicinity of BHs. The stable period of the QPO in ASASSN-14li suggests that it is tied to the physical properties (mass and spin) of the BH at the heart of the disruption.

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SUPPLEMENTARY MATERIALS

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Supplementary Text
Figs. S1 to S11
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Movie S1
Data S1 and S2

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